

KONGSBERG BRANNSTASJON, Norway

WMO: 014730

Lat: 59.6244N

Lon: 9.6375E

Elev: 173

StdP: 99.26

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -18.7	(c) -15.8	(d) -20.1	(e) 0.6	(f) -17.3	(g) -17.5	(h) 0.8	(i) -14.2	(j) 5.6	(k) 4.3	(l) 4.7	(m) 3.3	(n) 0.8	(o) 270	(p) 0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	27.4	16.5	25.3	15.7	23.2	14.7	17.8	24.0	16.9	22.7	16.0	21.2	2.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	11.5	19.0	14.8	10.7	18.2	13.9	10.1	17.6	50.7	24.0	48.1	22.8	45.5	21.3	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4.5	4.0	3.4	-20.4	30.1	3.1	1.6	-22.6	31.2	-24.4	32.1	-26.2	33.0	-28.4	34.1
			-19.9	19.4	2.1	1.2	-21.4	20.2	-22.7	20.9	-23.8	21.6	-25.4	22.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-4.3	-3.2	0.5	5.5	10.4	14.3	16.8	15.0	11.3	5.5	0.5	-3.6		
	DBStd	8.43	5.30	4.94	4.61	3.32	3.89	2.88	2.85	2.34	2.80	3.49	4.13	5.92		
	HDD10.0	2166	444	370	295	139	43	2	0	0	19	146	287	422		
	HDD18.3	4615	703	603	553	384	248	126	66	105	211	399	536	680		
	CDD10.0	623	0	0	1	5	54	131	212	157	58	6	0	0		
	CDD18.3	28	0	0	0	0	1	5	19	3	0	0	0	0		
	CDH23.3	412	0	0	0	0	37	97	238	38	2	0	0	0		
	CDH26.7	76	0	0	0	0	4	19	51	3	0	0	0	0		
(14) Wind	WSAvg	1.5	1.4	1.4	1.6	1.7	1.8	1.8	1.6	1.4	1.5	1.4	1.4	1.3		
(15) Precipitation	PrecAvg	968	77	47	49	59	65	84	91	103	103	110	106	70		
	PrecMax	1493	165	137	136	142	195	196	203	205	288	275	460	114		
	PrecMin	758	12	7	0	7	10	4	16	13	45	26	16	14		
	PrecStd	151	41	29	33	33	39	44	43	45	55	58	84	30		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	9.5	16.0	20.2	26.9	29.2	30.2	26.7	22.9	16.8	12.0	10.0		
		MCWB	4.7	4.8	8.1	10.9	14.8	15.9	17.6	16.7	15.4	12.7	8.6	7.0		
	2%	DB	6.1	6.9	12.5	17.9	23.8	26.2	28.1	24.1	20.3	14.6	10.1	7.3		
		MCWB	3.5	3.3	6.4	9.9	13.6	15.0	17.1	16.0	13.7	10.5	7.9	4.9		
	5%	DB	4.3	5.0	9.5	15.1	21.3	23.7	26.2	22.3	18.7	12.9	8.1	5.7		
		MCWB	2.0	2.2	4.5	8.3	12.5	13.8	16.6	15.1	12.8	9.6	6.3	3.5		
	10%	DB	2.6	3.1	7.3	12.7	18.6	21.3	23.9	20.6	16.9	11.2	6.4	4.0		
		MCWB	0.9	1.2	3.1	6.7	11.4	12.9	15.6	14.4	12.0	8.7	4.9	2.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	5.5	8.9	12.1	16.5	17.3	19.4	18.2	16.4	13.2	9.8	7.3		
		MCDB	7.5	8.8	15.6	18.4	23.4	24.6	25.6	23.1	20.8	15.8	11.0	9.7		
	2%	WB	4.0	3.8	6.8	10.3	15.1	16.1	18.2	17.3	14.9	11.4	8.1	5.4		
		MCDB	5.9	6.1	11.9	16.2	21.1	22.9	25.1	21.6	18.1	13.5	9.5	7.0		
	5%	WB	2.6	2.6	5.1	8.9	13.6	15.0	17.4	16.4	13.8	10.3	6.6	3.8		
		MCDB	4.0	4.6	8.9	14.2	19.2	21.7	24.0	20.5	17.0	12.1	8.1	5.4		
	10%	WB	1.2	1.4	3.6	7.5	12.0	14.0	16.5	15.5	12.9	9.1	5.1	2.3		
		MCDB	2.5	2.9	6.7	12.0	17.2	19.5	22.5	19.1	15.7	11.0	6.2	3.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	7.2	10.0	11.2	11.4	11.0	10.8	10.1	9.8	7.7	6.0	6.1		
		MCDBR	7.5	9.7	13.2	15.5	15.7	15.6	14.7	12.9	12.7	9.9	8.7	8.0		
	5% WB	MCWBR	5.9	6.7	8.0	8.4	7.2	7.1	6.4	6.2	7.2	6.5	6.9	6.3		
		MCWBR	7.0	8.4	12.6	14.2	12.6	12.9	12.6	10.4	10.4	8.2	8.0	7.5		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.291	0.317	0.360	0.358	0.348	0.368	0.362	0.331	0.317	0.272	0.226		
		taud	2.213	2.291	2.319	2.323	2.425	2.474	2.434	2.446	2.523	2.482	2.308	2.129		
	Ebn at Noon		556	724	818	840	871	886	856	833	806	698	529	437		
		Edh at Noon	41	67	88	105	102	99	101	93	73	56	38	29		
(44) All-Sky Solar Radiation	RadAvg		0.33	0.98	2.51	3.99	4.86	5.54	4.86	3.85	2.52	1.17	0.43	0.22		
	RadStd		0.07	0.21	0.22	0.56	0.58	0.45	0.64	0.29	0.35	0.23	0.11	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)		N/A	N/A	+1.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page